

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

May 2012

CE210 MICROPROCESSOR & COMPUTER ARCHITECTURE

Date: 19.05.2012, Saturday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions.

[11]

- [A] _____ Flip-flop is used for Storage Element in 8085. [1]
- [B] What is the need for chip selection in a microcomputer? [1]
- [C] What is use of following pins of 8085: HOLD,READY [1]
- [D] Why is the need to demultiplex the bus AD7-AD0? [1]
- [E] List the sequence of events that occur when 8085 reads data from 2000H memory location. [1]
- [F] Assume that memory location is 2075 has data byte 47H. Specify the contents of the address bus A15-AD8 and the multiplexed bus AD7-AD0 when the MPU asserts the Read signal. [2]
- [G] Write the differences between PUSH-POP and CALL-RET. [2]
- [H] Why are the program counter and stack pointer 16-bit registers? [2]

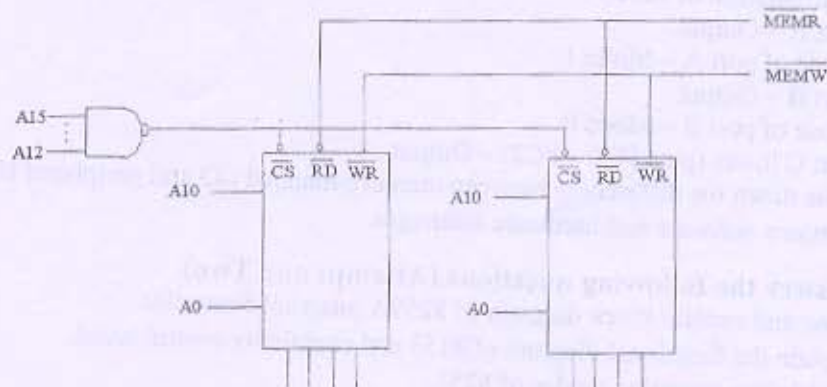
Q-2 Answer the following questions.

[12]

- [A] Write an assembly program to sort the given numbers in descending order. [4]
- [B] Write down various addressing modes of 8085 with example. [4]
- [C] What is flag? Which are the flags available in 8085 microprocessor? [4]

OR

- [A] Write a program to count continuously in hex from FFH to 00H in a system with 0.5 microsecond clock period. Use register C to setup a one millisecond delay between each count and display the numbers at one of the output ports. The T-states for various instructions are
DCR-4 MVI-7 JNZ-10/7 OUT-10 MOV-4 JMP-10 [4]
- [B] Explain following instructions: XCHG, SIM [4]
- [C] Specify memory map of the following diagram and explain significance of the don't care address line on memory addresses. [4]



Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] How does 8085 perform internal data operations? How many registers it has to perform internal data operations? Write functionality of each register.
- [B] What are Tri-State devices and why are they essential in a bus-oriented system?
- [C] Draw and explain Functional Block Diagram of 8085 microprocessor.
- [D] Draw and explain timing diagram of POP instruction.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] 8086 has _____ number of address lines and _____ number of data lines. [1]
- [B] What do you mean by Stack Creep? [1]
- [C] What do you mean by vector and non-vector interrupts? [1]
- [D] What is advantage of subroutine? [1]
- [E] Specify the number of times the following loops are executed. [1]

XRA A

LOOP: MVI B,64H

NOP

DCR B

JNZ LOOP

- [F] Specify the register contents and flag status(S, Z, CY) after the instruction OR A is executed. [2]

MVI A,A9H

MVI B,57H

ADD B

ORA A

- [G] Compare: JMP and CALL instructions in terms of function, addressing mode, type and affected flags. [2]
- [H] List various programmable peripheral devices which are compatible with intel microprocessors family. [2]

Q-5 Answer the following questions. [12]

- [A] Explain BSR mode of 8255. [4]
- [B] Write an assembly language program to multiply two 8-bit numbers using successive addition method. [4]
- [C] What are the characteristics of 80386 architecture? [4]

OR

- [A] What is Control Word of 8255? Determine the control word for the following configuration of 8255:- [4]

Port A – Output

Mode of port A – Mode 1

Port B – Output

Mode of port B – Mode 0

Port C lower (pins PC0 – PC2) – Output

- [B] Write down the differences between memory-mapped I/O and peripheral I/O. [4]
- [C] Compare software and hardware interrupts. [4]

Q-6 Answer the following questions.(Attempt any Two) [12]

- [A] Draw and explain block diagram of 8259A interrupt controller.
- [B] Explain the functional diagram of 8155 and explain its control word.
- [C] Explain the operating modes of 8253.
- [D] Compare: 8086, 80286, 80386.